A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 39

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
01.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2400 / 40 = 60 times **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Reserves:** the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) **1 Resource:** all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits **1** The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Lowering the cut-off ore grade includes deposits of lower purity in the reserves **1** Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	**1 mark each** 1 mark for each of the 3 points below. **1** The total quantity (stock / resource base) is far greater than could ever be used **1** But most is at purities too low or in locations too inaccessible for current economic exploitation **1** The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable – energy needs rise rapidly as ore grade falls **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** A waste is assumed to cause pollution until research shows it is unlikely to do so **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution **1** If polluters can escape their responsibilities / penalties are low, there is little incentive to comply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
02.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marine discharges where currents dilute and disperse emissions **1** Emissions downwind of urban areas **1** Not discharging waste onto permeable rock above an aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Effluent gases pass through a chamber containing electrically charged wires / plates **1** Smoke particles become (negatively) charged and are attracted to the oppositely charged plates where they collect / stick **1** Plates are shaken and the particles fall as fly ash which is collected (and may be used, eg in building materials) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **electrostatic precipitator** → charged plates that attract and collect particulates	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Effluent gases are forced to rotate / spin in a cylindrical chamber **1** The (denser) particles are thrown outwards to the chamber walls where they fall and collect – cleaned gases leave through a central pipe **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	1 mark for the point below. **1** The tar / volatile organic compounds that produce smoke on incomplete combustion are driven off, producing smokeless coal / coke **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Turbochargers deliver more air / oxygen to the combustion chamber **1** So combustion is more complete and fewer unburnt carbon particles / soot are produced (especially during acceleration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.5	**1 mark each** 1 mark for each of the 2 points below. **1** Smaller particles remain suspended in the atmosphere for longer so they are more likely to be inhaled **1** They penetrate deeper into the lungs / alveoli and are not cleared – may carry toxic / carcinogenic chemicals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
04.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up **1** High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) **1** Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) **1** Birds of prey are at the top of the food chain so receive the highest concentrations **1** Sub-lethal doses caused thin eggshells that broke / reduced breeding success – higher doses caused sterility or death **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **trophic level** → the feeding level in a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly **1 Disadvantage:** high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase) – chronic effects (memory loss, depression) – suspected carcinogens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	2
04.3	1 mark for the point below. **1** Pyrethroids are highly toxic to fish (even though mammal toxicity is low) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
04.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) **1** They are moderately persistent so remain in the soil long enough to be absorbed **1** Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats **1** So bees / pollinators are less exposed than when the crop is sprayed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Total = 42 + 26 + 31 + 9 + 22 = 130 kW **1** 31 / 130 x 100 = 23.8% (accept 24%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1 Cavity wall insulation:** the foam or fibre traps air in small pockets so convection in the cavity is prevented and the low thermal conductivity of still air reduces conduction through the wall **1 Low-emissivity double glazing:** the still air or inert gas between the panes conducts heat poorly, and the low-emissivity coating reflects infrared radiation back into the room instead of letting it radiate out **1 Heat exchanger:** outgoing stale warm air passes next to, but does not mix with, incoming cold fresh air, so most of the heat is transferred to the incoming air and the building can be ventilated without losing that heat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each extra layer saves less heat than the one before, because the layers already in place carry most of the temperature difference, so the saving per unit thickness falls **1** Beyond a certain thickness the cost and the embodied energy of making and fitting more insulation are never repaid by the energy it saves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
05.4	**1 mark each** 1 mark for each of the 3 points below. **1** A large power station runs at higher temperatures and pressures and uses large efficient turbines, so it converts a greater proportion of the fuel energy into electricity **1** The heat rejected by a large station is low-grade and the station is usually too far from any user, so it is lost to a river or cooling tower **1** A CHP unit is in the building that needs the heat, so the heat rejected by its generator is piped straight to the heating and hot water, and 70 - 80% of the fuel energy is used rather than about 35% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **combined heat and power** → using the waste heat from generation for heating	3

Do not write outside the box

Question	Marking guidance	Marks
05.5	**Any 5 of 6** 1 mark each for any 5 of the 6 points below, in any order, maximum 5. **1** Occupancy sensors and programmable thermostats so that lighting and heating run only where and when people are present **1** LED lighting, which converts a far greater proportion of the electricity into light and much less into waste heat than filament lamps **1** Large south-facing windows and a thermally massive structure to make use of passive solar gain, with shading to prevent overheating in summer **1** Heating water on demand at the point of use, so no energy is wasted keeping a tank hot and none is lost in the pipe run **1** Choosing materials with low embodied energy, and re-using the existing structure rather than demolishing and rebuilding **1 Changes in behaviour:** turning equipment off rather than leaving it on standby, closing doors, and setting the thermostat a degree lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-235; AQA spec 3.3.4.2.2-3.3.4.2.3.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 B:** Oxygen	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) **1** There is still a lot that is not understood about how atmospheric processes work **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** The surface warmed in the tropics heats the air above – warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes **1** Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 5 points below. **1** Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated **1** Same slope / gradient – simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time **1** Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it **1** Repeat each treatment several times and calculate means (and standard deviation) **1** Plot soil loss against cover – statistical test (Spearman's rank / t-test) for significance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean • **Spearman** → a test for correlation between two sets of ranked data • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	5
07.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Rainfall intensity / raindrop size / height **1** Soil texture, compaction, organic matter, depth **1** Permeability of the material beneath **1** Gradient **1** Root binding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Permeability** → how easily water passes through / lets water through **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** All variables except the independent variable can be standardised in the laboratory – in the field uncontrollable variables (weather) must be measured instead **1** Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **valid** → actually measures what it is meant to measure	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1
08.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** The amount of energy contained in a given mass of the resource (or harnessed per kg of equipment for renewables) **1** High energy density means smaller quantities are needed so storage and transport are easier and cheaper (eg 75 t of aviation fuel carries a 400 t aircraft 5600 km) **1** High energy density fuels reach high temperatures needed for smelting / steam turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 500 000 / 28 = 17 857 kg **1** = 17.9 tonnes (accept 18; book quotes 13 t for 0.7% enriched fuel) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** The electricity they generate can power high-energy-requirement technologies **1** Electricity can be converted to high energy density secondary fuels, eg hydrogen by electrolysis, or stored (batteries, pumped storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1 Intermittent:** not available all the time, eg tidal (but the times are predictable so alternatives can be planned) **1 Unpredictable:** intermittent and the timing / amount cannot be forecast accurately, eg wind, solar **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • Any appropriate named example, including ones that are not listed here. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2
09.5	1 mark for the point below. **1** Gaseous natural gas has a very low energy density by volume – LNG at -160 °C has 160 times the energy density so a ship can carry a worthwhile amount **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-178; AQA spec 3.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** No reservoir range = 3.9 - 0.7 = 3.2 m – with reservoir = 2.7 - 1.7 = 1.0 m **1** Reduction = 2.2 m **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream **1** During dry periods the dam is opened to release water and maintain river levels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** Deforestation / urbanisation reduce infiltration and interception **1** So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Site factors:** topography - narrow exit from a large deep basin so a small dam holds a large volume – impermeable, strong bedrock without faults or seismic activity – large catchment with regular high rainfall and low evaporation – catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation – forest / vegetation in the flooded area decaying anaerobically releasing methane – access for workers and materials and distance to demand – value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • **Alternatives for supply:** rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • **Conservation strategies:** low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • **Evaluation:** reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane – conservation is cheap and quick but limited by behaviour – transfers create conflicts between regions – desalination has high energy / carbon cost – combination needed as demand grows • Justified conclusion	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **impermeable** → does not let water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	
	Model answer - Level 3 (8 marks)	
	A reservoir site is judged first on topography: a narrow exit from a large, deep basin lets a small, cheap dam hold a large volume. The bedrock must be impermeable so water is not lost and strong enough, without faults or seismic activity, to carry the dam. The catchment should be large with regular, high rainfall and low evaporation, and its land uses should not pose a pollution risk from industry or pesticides or cause the soil erosion that would silt the reservoir. Flooding forest or vegetation causes anaerobic decay and methane. Access for workers and materials, distance to the area of demand, and the value of the land lost - urban areas and nature reserves protected, farmland less so - complete a loss-benefit analysis that different countries weigh differently.	
	Supply can also be increased without new reservoirs: rainwater collection in cities, estuary barrages, desalination where freshwater is genuinely inadequate, newly found aquifers in North Africa and Kenya, inter-basin transfers as in China and from Wales to England, artificial recharge of aquifers with wet-season surplus, river-regulation reservoirs and afforestation to even out flow. Demand can be cut by low-water appliances, xeriscaping, drip irrigation, grey-water re-use, metering and repairing the leaks that lose about a fifth of UK supply, and treatment lets contaminated sources be re-used.	
	Reservoirs give large, reliable supplies and flood control but flood land and habitats and release methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination carries a high energy and carbon cost. As demand grows, the sustainable answer is conservation first, then a combination of sources matched to each region.	
	Weaker answer - Level 2 (4 marks)	
	A reservoir needs a large catchment area with lots of rain, impermeable rock so the water does not drain away, and a narrow valley so the dam is small. It should be near the people who need the water and not flood important land. Other ways to get water include desalination, which is expensive, and collecting rainwater. Water can be conserved by fixing leaks, using water meters and having low-flush toilets. All of these help make water supplies more sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 lists correct site factors but without the reasons (why impermeable, why the valley shape matters, why land use in the catchment matters, methane from flooded vegetation).	
	• Level 3 covers far more alternatives (aquifer recharge, river regulation, transfers, afforestation) and evaluates each; Level 2 evaluates only desalination.	
	• Level 3 ends with a prioritised judgement; Level 2 ends with 'all of these help'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Historical data limitations – proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • **Satellite sensors:** visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring – passing through cloud • **Sensor carriers:** ROVs / drones, AUVs, balloons, aircraft, animals – data recorders for abiotic factors – GPS tracking of floats and sampling sites – Argo floats and ocean monitoring • Climate models and their assumptions – monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • **Evaluation:** improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	

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